

Question Number	Answer	Additional guidance	Mark
7(a)	correct calculation of magnification	(40 ÷ 4.4 = 9.091) (x) 9.1 reject units	(1)

Question Number	Answer	Additional guidance	Mark
7(b)	The only correct answer is B 0.20% <i>A is not correct because 0.20% of infected people died</i> <i>C is not correct because that is the percentage of infected people who survived</i> <i>D is not correct because 0.20% of infected people died</i>		(1)

Question Number	Answer	Additional guidance	Mark
7(c)(i)	produce ribosomal subunits	Accept produce ribosomes ignore contains genetic material	(1)

Question Number	Answer	Additional guidance	Mark
7(c)(ii)	A description that includes three from the following points: • plant cell would have {starch (grain) / amyloplast} (instead of glycogen granule) (1) • plant cell would have cell wall made from cellulose (not chitin) (1) • plant cell would contain {plasmodesmata / one nucleus} (whereas this cell does not) (1) • the plant cell vacuole would be larger (1)	ignore chloroplast accept plant cells do not have glycogen granules accept plant cell would have lignin accept plant cells do not have chitin cell wall ignore pores	(3)

Question Number	Answer
7 (d)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. • description of the table data e.g. comparison of numbers of mosquitoes {killed / survived} • consideration of GM fungus being most effective treatment • consideration that most mosquitoes were resistant to the insecticide used / some were not resistant to insecticide and were killed / inheritance of resistance • consideration that resistance arose due to mutation / natural selection / genetic variation in population • as gene is only {switched on / expressed} when the fungus has infected an <i>Anopheles</i> mosquito • the toxin is only produced when the fungus has infected an <i>Anopheles</i> mosquito • consideration of protein synthesis / detail of the role of rER and Golgi apparatus • evaluation of methodology / limitations of method e.g. mosquitoes escaping, small sample size • consideration of wider effects on biodiversity e.g. toxin will not poison other insects / humans / organisms whereas insecticides would kill other insects / effect on food chains / biodiversity

	transfer of (spider venom) gene to other fungal species			(6)
			Additional guidance	
Level 0	0	No awardable content		
Level 1	1-2	Limited scientific judgment made with a few strengths / weaknesses identified. A conclusion may be attempted, demonstrating isolated elements of biological knowledge and understanding but with limited evidence to support the judgement being made.		1 mark – description of results using table data. 2 marks consideration of one from: consideration of most effective treatment resistance / inheritance of resistance mutation gene expression protein synthesis limitations of method biodiversity and transfer of gene to other fungi
Level 2	3-4	A scientific judgment is made through the application of relevant evidence, with strengths / weaknesses identified. A conclusion is made, demonstrating linkages to elements of biological knowledge and understanding, with occasional evidence to support the judgement being made.		all level one plus: 3 marks – two from list 4 marks – three from list consideration of: consideration of most effective treatment resistance / inheritance of resistance mutation gene expression protein synthesis limitations of method

			biodiversity and transfer of gene to other fungi
Level 3	5-6	A scientific judgment is made which is supported throughout by sustained application of relevant evidence from the analysis and interpretation of the scientific information. A conclusion is made, demonstrating sustained linkages to biological knowledge and understanding, with evidence to support the judgement being made.	all level 2 content 5 marks – consideration of four from the list 6 marks – consideration of five from the list consideration of most effective treatment resistance / inheritance of resistance mutation gene expression protein synthesis limitations of method biodiversity and transfer of gene to other fungi